Algorithmic Bias Mitigation on Adult Census Income

An in-processing study reproducing Agarwal et al. (2018), *A Reductions Approach to Fair Classification*, and measuring three consequences its framing does not surface.

Summary

The base paper's method works exactly as claimed. Wrapped around a decision tree, its exponentiated-gradient reduction takes demographic parity difference from 0.1613 to 0.0187 — an 88% reduction — for 1.5 accuracy points, without modifying a single row of training data. Every claim the paper makes held up under independent reimplementation.

That confirmation is the first half of this report. The second half asks four questions the paper's framing does not: **who** was moved to satisfy the constraint, whether the total number of favourable decisions fell, what the method's randomization costs an individual, and what the constrained model actually leans on. The answers are less comfortable than the headline metric.

> **Every mitigation studied closed the fairness gap primarily by withdrawing favourable decisions from the advantaged group rather than extending them to the disadvantaged one, reducing the total number of approvals by 8–22%. And the two methods with the best parity scores rely on proxies for the protected attribute *more* than the unmitigated baseline does.**

Contributions

- **Reproduction** of the base paper's four deliverables on an independent implementation, over five random seeds.
- **An ablation** of six in-processing methods under identical conditions, two of them implemented from their source papers rather than from a library, and verified against degenerate cases.
- **An incidence analysis** (beyond the specification) decomposing each closed fairness gap into the part paid by the advantaged group losing ground and the part gained by the disadvantaged group, in rates and in people. The audit axes are Ferry et al.'s (2023); what is ours is the measurement on these six methods.
- **A proxy-reliance analysis** using SHAP, which refutes this project's own stated prediction.
- **An intersectional analysis** (beyond the specification) at Sex × Race, reproducing Kearns et al. (2018) and Maheshwari et al. (2023), with a measurement-reliability treatment — the part that is ours — showing that half the subgroups on this dataset cannot support the estimates commonly published for them.

1. Setup

UCI Adult Census Income, 45,222 rows after listwise deletion. Task: predict whether income exceeds $50K. Protected attribute: `sex`. Base rates differ sharply — 31.25% of men in the data earn above the threshold against 11.36% of women — and that 2.75× ratio is the entire source of the bias. Nothing in the algorithm is malicious; empirical risk minimisation reproduces the gap because reproducing it minimises error.

The protected attribute is removed from the feature matrix. No model in this study can read `sex` directly. This is fairness through unawareness, included precisely because it does not work: the models reconstruct the disparity from proxies, and section 6 measures which ones.

Splits are stratified on the `(sex, income)` interaction rather than the label alone, so the four cells the fairness metrics are computed from keep a stable size across seeds. Without this, sampling noise in the smallest cell appears as model instability — which would have confounded the stability finding in section 4. All three fairness metrics are implemented from their definitions and cross-checked against `fairlearn` on every run; rates with an empty denominator return NaN rather than 0, so an undefined rate can never masquerade as a perfectly fair one.

2. Baseline

Base classifier	Accuracy	DP diff	EO diff	Disparate impact
Decision tree	0.8517 ± 0.0030	0.1613 ± 0.0125	0.0831 ± 0.0234	0.3086 ± 0.0315
Logistic regression	0.8469 ± 0.0033	0.1861 ± 0.0083	0.0949 ± 0.0295	0.2996 ± 0.0296

Disparate impact of 0.29–0.31 means women are selected at under a third of the male rate. The US EEOC four-fifths rule flags anything below 0.80; this is off by nearly a factor of three. Accuracy sits near 85% throughout — a reader monitoring only accuracy would conclude the pipeline is healthy, which is the ordinary case.

3. Reproducing the base paper

Agarwal et al. rewrite the constrained problem $\min_h E[1\{h(x)\neq y\}]$ subject to $\phi(h) \leq \varepsilon$ as a two-player zero-sum game, $\min_h \max_\lambda \text{error}(h) + \lambda^T(\phi(h) - \varepsilon)$. In practice this reduces to repeatedly reweighting the training examples and refitting the base classifier, so the classifier is a black box and the data is never touched. Decision tree base, $\varepsilon = 0.01$, 5 seeds.

Method	Accuracy	DP diff	EO diff	Disparate impact
Baseline tree	0.8517 ± 0.0030	0.1613 ± 0.0125	0.0831 ± 0.0234	0.3086 ± 0.0315
ExpGrad — Demographic Parity	0.8364 ± 0.0018	0.0187 ± 0.0069	0.2773 ± 0.0171	0.8834 ± 0.0423
ExpGrad — Equalized Odds	0.8471 ± 0.0030	0.1125 ± 0.0139	0.0355 ± 0.0242	0.4573 ± 0.0486

The method works. Under the parity constraint the violation falls 88% for 1.5 accuracy points, and disparate impact moves from failing the four-fifths rule by a factor of three to comfortably passing it. Under the equalized-odds constraint, EO difference falls 57% for half an accuracy point. Each constraint improves the metric it was given and only that one — correct behaviour, and the seed of the finding in section 4.

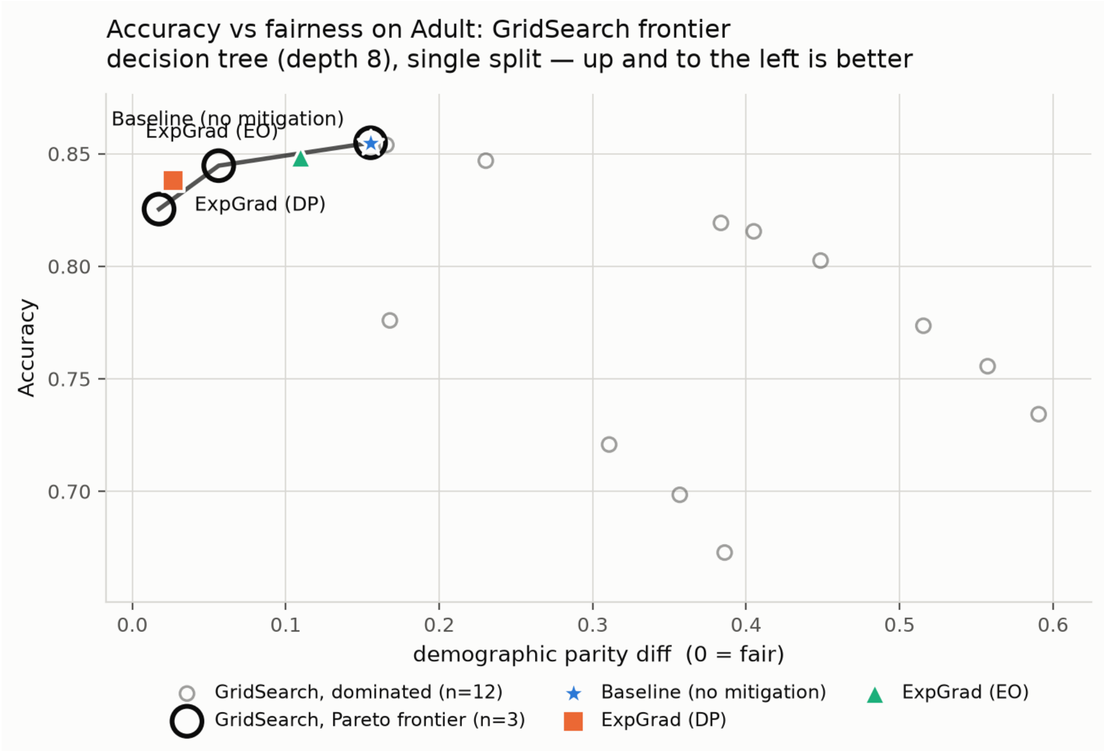

Figure 1. GridSearch sweep over 15 λ values (base-paper deliverable 4). Open rings mark the non-dominated frontier; dominated points are shown rather than hidden, because a grid producing many of them is evidence the trade-off is less smooth than a frontier-only plot implies.

4. Ablation: six methods, one table

Data, base classifier, split and metrics held fixed; only the mitigation varies. All six rows use logistic regression, including the reductions rows that section 3 runs on a tree: Prejudice Remover adds a term to a likelihood and Adversarial Debiasing needs gradients to flow from an adversary into the predictor, so neither can wrap a tree. Reporting a tree for some rows and a linear model for others would vary the hypothesis class and the mitigation at once.

Method	Accuracy	DP diff	EO diff	Disparate impact
Baseline (no mitigation)	0.8469 ± 0.0033	0.1861 ± 0.0083	0.0949 ± 0.0295	0.2996 ± 0.0296
Exponentiated Gradient (DP)	0.8282 ± 0.0023	0.0178 ± 0.0091	0.2802 ± 0.0190	0.8951 ± 0.0516
Exponentiated Gradient (EO)	0.8361 ± 0.0012	0.1072 ± 0.0101	0.0320 ± 0.0144	0.5214 ± 0.0389
GridSearch (DP)	0.8265 ± 0.0070	0.0150 ± 0.0190	0.3040 ± 0.0460	0.9859 ± 0.1377
Prejudice Remover	0.8365 ± 0.0020	0.0646 ± 0.0085	0.1884 ± 0.0247	0.6860 ± 0.0373
Adversarial Debiasing	0.8260 ± 0.0011	0.0202 ± 0.0093	0.2504 ± 0.0148	0.8894 ± 0.0498

Prejudice Remover and Adversarial Debiasing were implemented from Kamishima et al. (2012) and Zhang et al. (2018) in PyTorch rather than taken from `aif360`, whose Prejudice Remover shells out to the original script through temporary files and whose Adversarial Debiasing requires a TensorFlow-1 compatibility chain. The cost of implementing from scratch is that the implementations might be wrong, so they are tested against degenerate cases: with $\eta = 0$ the prejudice penalty vanishes and the method must reduce to per-group logistic regression, which it reproduces to **99.88%** prediction agreement. Both fairness knobs are checked for monotonicity — the check that would catch a sign error, the failure mode where a mitigation quietly increases disparity while the metrics still look plausible.

4.1 Which method gives the best value?

GridSearch-DP reaches the lowest absolute violation (0.0150), but the best *exchange rate* belongs to Prejudice Remover, which buys 11.7 parity points per accuracy point against ExpGrad-DP's 9.0 — roughly 30% better. It stops at DP 0.065 rather than pushing to 0.015, because η is a penalty weight rather than a constraint and has no target it is obliged to hit. Its efficiency win also carries a caveat the reductions rows do not: it fits one weight vector per group, so it requires `sex` at *prediction* time. Two applicants identical on every feature but differing in sex are scored by different weight vectors — disparate treatment in the legal sense, arrived at while trying to reduce disparate impact.

4.2 Which method is most stable? (Our prediction was inverted.)

Method	Accuracy std	DP std	Disparate impact std
Adversarial Debiasing	0.0011	0.0093	0.0498
Exponentiated Gradient (DP)	0.0023	0.0091	0.0516
GridSearch (DP)	0.0070	0.0190	0.1377

The project's own specification predicted that adversarial training would be the higher-variance approach. It is the **most** stable method in the study. GridSearch — a deterministic sweep with no adversary, no minibatching and no randomness in the procedure — is the least, by 6× on accuracy and 3× on disparate impact. The variance is measured across seeds, and a seed changes the train/test split: GridSearch takes a discrete argmax over a coarse 15-point λ grid, so a small change in the data flips which grid point wins and the answer moves discontinuously. Adversarial Debiasing has no selection step. **The instability came from model selection, not from stochastic training.**

4.3 Does the ranking change with the metric?

It inverts. Ranked by demographic parity the order runs GridSearch → ExpGrad-DP → Adversarial → Prejudice Remover → ExpGrad-EO → Baseline. Ranked by equalized odds it becomes ExpGrad-EO → **Baseline** → Prejudice Remover → Adversarial → ExpGrad-DP → GridSearch.

Four of the five mitigations are worse than doing nothing on equalized odds, and the unmitigated baseline ranks second of six. An engineer deploying GridSearch-DP on its parity score would take equalized odds from 0.095 to 0.304 — 3.2× worse than no mitigation — while the dashboard showed green.

This is not a defect in any implementation. It is the impossibility result (Kleinberg et al. 2016; Chouldechova 2017) appearing in practice: when base rates differ across groups, demographic parity and equalized odds cannot both hold. “We made the model fair” is not a well-formed claim on this data.

5. Who pays for the fairness fix?

This section and section 7 are outside the original specification. The ablation table reports that ExpGrad-DP took demographic parity from 0.186 to 0.018. That single number is compatible with two opposite stories: the disadvantaged group's selection rate rose to meet the advantaged group's (*levelling up*, nobody made worse off), or the advantaged group's fell to meet theirs (*levelling down*). Same metric, opposite ethics. Mittelstadt, Wachter & Russell (2023) argue the second is the common case and that gap-only reporting conceals it.

Writing the signed gap as gap = $r_{priv} - r_{unpriv}$, the change decomposes exactly:

```
closure = (r_priv_before − r_priv_after) + (r_unpriv_after − r_unpriv_before)
          \____ privileged loss ____/   \____ unprivileged gain ____/
```

The two terms sum to the closure identically, so their ratio is a well-defined share. The identity is verified over 200 random cases, and a gap that *widened* returns NaN rather than a plausible-looking number (8/8 tests).

Method	Share paid by privileged (rates)	Share paid by privileged (people)	Lost per gained	Change in total approvals
Exponentiated Gradient (DP)	0.575	0.742	2.68	-20.5%
GridSearch (DP)	0.575	0.743	2.69	-22.1%
Adversarial Debiasing	0.508	0.700	1.96	-14.8%
Prejudice Remover	0.498	0.673	2.04	-10.2%
Exponentiated Gradient (EO)	0.531	0.660	1.92	-7.9%

Measured in rates every method looks even-handed (0.50–0.58 of the closure paid by the privileged group). **Measured in people they are lopsided** (0.66–0.74). Both are correct; they answer different questions. The rate decomposition is population-size blind, and the privileged group here is 2.1× larger, so equal rate movement is very unequal headcount — and headcount is what a person subject to the system experiences.

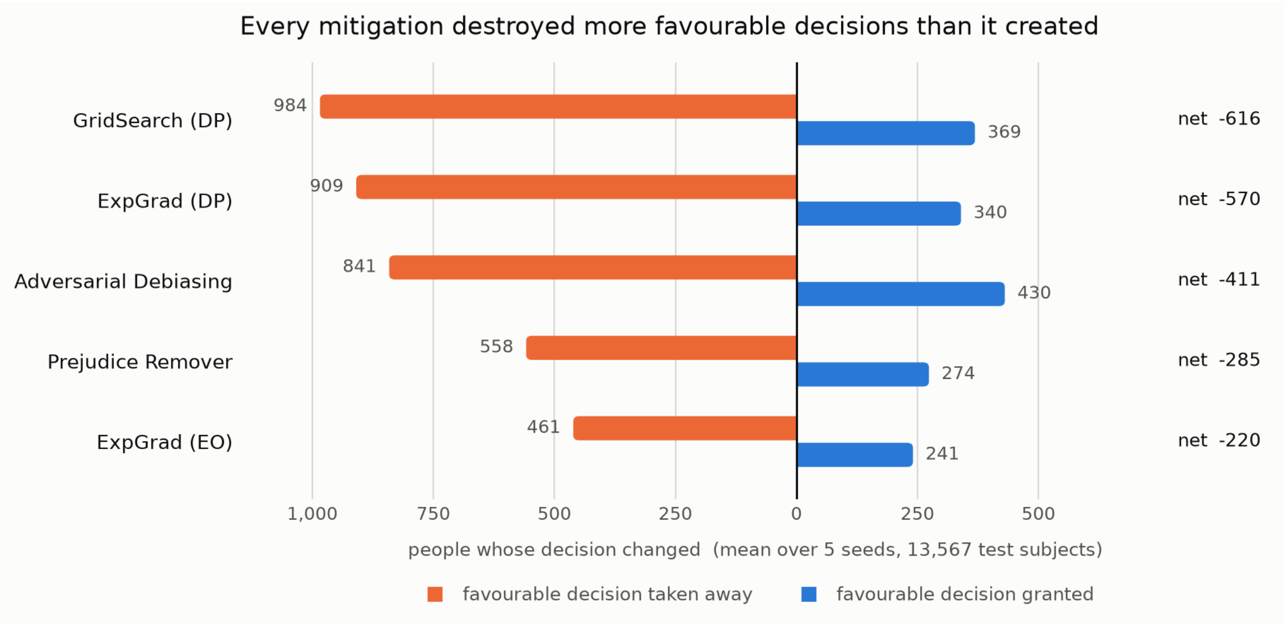

Figure 2. Favourable decisions withdrawn and granted, in people, mean over five seeds. Not one method closed the gap primarily by extending favourable decisions to the disadvantaged group.

ExpGrad-DP took approval away from 909 men so that 316 women could gain it. Every method reduced the total number of favourable decisions — by 7.9% to 22.1%. Demographic parity constrains a ratio, not a total, and the cheapest way to fix a ratio is usually to shrink the numerator.

5.1 How much of the effect is not fairness at all

`ExponentiatedGradient` returns a distribution over classifiers and samples one at predict time, so some subjects receive a different decision on every call regardless of any constraint. The **arbitrariness floor** measures this by drawing twice from the same fitted model.

Method	Subjects whose decision changed	Arbitrariness floor	Share of the effect that is noise
Exponentiated Gradient (EO)	5.17%	3.18%	62%
Exponentiated Gradient (DP)	9.21%	1.18%	13%
GridSearch (DP)	9.97%	0.00%	0%
Adversarial Debiasing	9.37%	0.00%	0%

About 62% of the individual decisions ExpGrad-EO changes are re-sampling, not fairness. Reporting its churn as the constraint's effect would be wrong by a factor of two and a half. The same applicant applying twice can receive different answers, for reasons unconnected to fairness or to their application. This is not a bug — the paper is explicit that the output is randomized, and the randomization is what makes the theory work — but it is a cost that falls on individuals and appears in no group metric.

6. What does the mitigated model actually use?

The specification listed SHAP as a stretch goal, expecting it to “show reliance on `sex` and its proxies **shrinking** post-mitigation”. It does not. For the two best parity methods it grows.

`sex` is absent from the feature matrix, so any sex-related reliance is proxy reliance by construction. The proxies examined are `relationship` (whose levels on Adult are literally Husband and Wife), `marital-status`, `occupation` and `hours-per-week`. Attributions are aggregated back to source features — exact, since Shapley values are additive — and reported as a share of each model's total attribution mass, because the six models emit scores on different scales. Five of seven explainers are exact linear SHAP; the two randomized ensembles are sampled, and that is flagged rather than hidden.

Model	Proxy reliance	vs baseline	relationship alone	SHAP
Baseline (no mitigation)	0.545 ± 0.006	—	0.075	exact
Exponentiated Gradient (DP)	0.586 ± 0.015	+7.6%	0.189	sampled
GridSearch (DP)	0.586 ± 0.023	+7.5%	0.158	exact
Prejudice Remover [Male]	0.505 ± 0.007	-7.3%	0.071	exact
Prejudice Remover [Female]	0.596 ± 0.077	+9.4%	0.179	exact
Exponentiated Gradient (EO)	0.475 ± 0.022	-12.8%	0.082	sampled
Adversarial Debiasing	0.481 ± 0.005	-11.7%	0.098	exact

Mean over 5 seed(s).

> **The two methods with the best demographic-parity scores lean on sex proxies harder than the model with no fairness fix, and both roughly doubled their use of *relationship*. To equalise selection rates across sex while forbidden to read sex, a model must first infer sex in order to compensate — the constraint gives it a reason to become a better sex-detector.**

Two things make this a mechanism rather than an artifact. First, ExpGrad's figure is sampled but GridSearch's is exact linear SHAP, and the two agree in direction and magnitude despite sharing only the constraint. Second, the one method that substantially *reduces* proxy reliance is Adversarial Debiasing — precisely the method whose objective is to make the output uninformative about sex. Theory predicts the ranking and the measurement matches it.

The practical consequence: the reduction's operational selling point is that the deployed classifier does not require `sex`. That is true, and it is a claim about the API, not about the model's reasoning. If a regulator asks whether a model treats men and women differently, the absence of `sex` from the input schema is not an answer.

7. Intersectional: Sex × Race

Every method above constrains one binary attribute, which is the field's default and has a known failure mode: constraints on marginals can hold while the cells inside them stay unfair (Kearns et al. 2018), and the harm at the intersection tends to go unnoticed in a model's overall performance figures (Maheshwari et al. 2023). Three arms on identical splits, 3 seeds. Fairlearn's reductions accept a multi-valued sensitive feature directly, so arm 3 is not a new algorithm — the contribution is the measurement.

7.1 The measurement problem comes first

Sex × Race splits the test set into ten subgroups spanning three orders of magnitude, from Male × White at 8,110 people down to Female × Other at 21. **Five cannot support a rate estimate.** Female × Other has no positive labels at all, so its true-positive rate is undefined by division, not by convention. A ten-cell heatmap printed without that caveat reports sampling noise as discrimination.

Arm	Gap over all 10 subgroups	Gap over the 5 measurable	Inflation
No constraint	0.3541	0.3145	0.0396
ExpGrad-DP on Sex	0.2207	0.1776	0.0431
ExpGrad-DP on Sex × Race	0.1606	0.0481	0.1124

For the intersectionally-constrained arm, 70% of the apparent gap is contributed by subgroups too small to measure. Every gap quoted below is the reliable one, and the widest reliable gap in each arm was checked for overlapping 95% Wilson intervals — in all three arms they do not overlap, so these are real differences.

7.2 Bias hides at the intersection

Arm	Accuracy	Sex gap	Sex × Race gap	Worst-off subgroup
No constraint	0.8465	0.1897	0.3145	Female x Black
ExpGrad-DP on Sex	0.8295	0.0197	0.1776	Male x Black
ExpGrad-DP on Sex × Race	0.8129	0.0164	0.0481	Female x White

Constraining on sex takes the sex gap to 0.0197 — essentially solved, and the number the ablation table reports. At the intersection the same model still carries a gap of 0.1776, 9× larger than the number on its own dashboard. Black men are selected at 9.2% against Asian men at 30.7%.

The sex constraint moved the worst-off subgroup from Black women to Black men. Before mitigation Black women were at the bottom; the constraint raised women's selection rates across the board, including theirs, but Black men were protected by no constraint at all and became the residual. Nothing was done to them — the constraint simply had nothing to say about them. A group can be made worst-off by a fairness intervention without ever appearing in it.

Constraining on the intersection instead cuts the reliable gap by 73% for 1.7 additional accuracy points, and is the only arm in this study that **lifts the floor**: the worst-off subgroup's selection rate triples, 0.052 → 0.157. The mechanism differs from section 5 because with ten groups rather than two, the constraint cannot satisfy itself by trimming one large group's numerator without opening gaps against the eight it is not trimming.

8. Two ways out that do not work

Sections 5 and 6 invite two obvious responses. If the model leans on `relationship`, delete it. If the constraint levels down, loosen it. Both were tested; neither works.

8.1 Deleting the leaky feature

> **This sub-section is deliberately *not* in-processing. Every method in sections 3 and 4 changes only the objective; deleting features is pre-processing, and it is run here as a negative control rather than as a proposed mitigation. Its result defends this project's in-processing scope rather than departing from it: no row of the ablation table comes from here.**

`FairnessDataset.attribute_leakage()` trains a probe to predict `sex` from the *remaining* features and reports ROC AUC. This is the direct measurement SHAP can only approach sideways: if the attribute is still recoverable after a deletion, the column went and the information stayed. Features are removed cumulatively, most sex-determining first. 3 seeds.

Features removed	Sex leakage (AUC)	Baseline acc	Baseline DP	ExpGrad acc	ExpGrad DP
none removed	0.934	0.8465	0.1897	0.8295	0.0197
–relationship	0.868	0.8463	0.2054	0.8026	0.0172
–relationship, marital-status	0.803	0.8171	0.1014	0.7946	0.0164
–relationship, marital-status, occupation	0.691	0.8133	0.0932	0.7852	0.0182
–relationship, marital-status, occupation, hours-per-week	0.625	0.8080	0.0761	0.7863	0.0186

Chance leakage is 0.500; the majority-class rate is 0.675.

Removing `relationship` — which determines sex outright for 45.9% of the dataset — moves leakage only from **0.934 to 0.868**. The information was never in that column alone. Suppressing the leak takes four deletions out of eleven features, and at that point the probe's accuracy (0.676) is indistinguishable from the majority-class rate (0.675).

Worse, the deletion **made the unmitigated model more unfair**: demographic parity difference rose from 0.190 to 0.205, with attribution simply relocating to `marital-status`, which remained the top feature in both rounds. The intervention that feels most obviously correct moved the metric the wrong way.

> **Feature deletion is strictly dominated. Removing four features yields DP 0.076 at 80.8% accuracy; the constraint on the untouched feature set yields DP 0.020 at 83.0%. Worse on both axes — and mitigation becomes more expensive afterwards, ExpGrad's accuracy falling from 0.830 to 0.803 as soon as *relationship* is gone.**

8.2 Loosening the constraint

Every result in section 5 used $\varepsilon = 0.01$, which leaves almost no slack. If levelling down is an artifact of an unusually tight constraint, the fix is to loosen it. The sweep below tests that. The baseline gap is 0.190, so $\varepsilon \geq 0.15$ is non-binding and must reproduce the baseline exactly — which it does, and that is the check that the parameter now controls what it claims to.

ε	Accuracy	DP diff	Share paid by advantaged (people)	Lost per gained	Change in total approvals
0.005	0.8284	0.0122	0.746	2.74	-22.0%
0.01	0.8295	0.0197	0.746	2.74	-21.1%
0.02	0.8324	0.0342	0.744	2.73	-19.1%
0.05	0.8388	0.0762	0.739	2.69	-13.6%
0.1	0.8457	0.1557	0.777	3.41	-5.0%
0.15	0.8465	0.1897	not binding	—	+0.0%
0.2	0.8465	0.1897	not binding	—	+0.0%

The share does not move. Across the entire binding range the advantaged group bears 0.739–0.777 of the closure measured in people. Total approvals fall at every binding ε, and closing a fixed amount of gap costs a near-constant number of approvals — roughly 120 to 146 per unit — whatever ε is set to. ε is a dial on how much fairness is bought, not on how it is bought.

At the loosest binding setting the split is in fact the *most* lopsided per unit of work, at 3.41 favourable decisions destroyed per one created. The likely reason is that a small correction is cheapest to make by trimming the largest group's selection rate slightly, which touches many people because that group is 2.1× larger.

A bug was found here before the finding was. The first run of this sweep returned identical numbers at every ε, including where the constraint cannot bind. `fairlearn`'s `DemographicParity()`, constructed with no arguments, pins its violation bound at the library default of 0.01, and the `eps` argument on `ExponentiatedGradient` sets only the Lagrange multiplier bound B = 1/ε. The sweep had been varying a parameter that never touched the constraint. No previously reported result changes — every other experiment used ε = 0.01, which coincides with that default, verified by reproducing seed 0's gap of 0.0282 exactly after the fix.

9. Verdict against the base paper

The distinction between extending a paper and refuting it is easy to blur and worth keeping sharp.

Confirmed

- The reduction drives the fairness violation toward ε for a modest accuracy cost — 88% reduction for 1.5 accuracy points.
- The base classifier is a black box: the identical wrapper worked unmodified on a decision tree and on logistic regression.
- The training data is never modified; reweighting is internal to the objective.
- GridSearch traces a usable accuracy-versus-fairness frontier.
- The output is a randomized classifier — confirmed, and quantified in 5.1.

Extended (outside the paper's frame, not against it)

Agarwal et al. is a paper about feasibility and optimality: given a constraint, find the most accurate classifier satisfying it. Within that frame every result here is a success. The findings in sections 5–8 are questions the frame does not ask — who was moved, whether the total fell, what randomization costs an individual, what the model leans on, what happens one level below the constrained attribute, and whether either obvious escape route works.

Refuted

Nothing in the base paper. The two refuted predictions are this project's own: that adversarial training would be the higher-variance approach (it is the most stable, and GridSearch is the least), and that SHAP would show proxy reliance shrinking after mitigation (it grew). One caution does touch the paper indirectly: GridSearch is presented there as the practical alternative for binary protected attributes, and its frontier is its selling point — this study suggests reporting its variance across resamples alongside that frontier, because the frontier looks smooth while the selection on it is fragile.

What a practitioner should take from this

- **Use the reduction.** It works, it is cheap, it wraps anything, and it leaves the data alone.
- **Do not report the gap alone.** “DP fell from 0.186 to 0.018” and “909 men lost approval so 316 women could gain it, and 570 fewer people were approved overall” are both true; only the second describes what happened.
- **Say which metric.** Four of five mitigations here made equalized odds worse than no mitigation while improving demographic parity.
- **Do not treat the absence of the protected attribute from the feature list as evidence of anything.** Tightening the constraint made the model lean on proxies harder.
- **Quantify the arbitrariness floor** before attributing individual-level changes to a fairness intervention.
- **Do not delete the proxy.** It leaves the information in place, can make the unmitigated model more biased, and loses to the constraint on fairness and accuracy at the same time.
- **Do not expect a looser constraint to be gentler in kind.** It is gentler only in degree, and per unit of gap closed it is not even that.
- **Check one level below the attribute you constrained** — and report subgroup sizes, so a reader can tell a finding from a small sample.

Limitations

- **One dataset.** Every finding is a statement about Adult, which is a 1994 census extract with known idiosyncrasies. Whether these results are properties of constraint-based reductions or of this dataset is not established here, and replication on ACS Income is the obvious next step.

- Race is used as recorded in the source data: five administrative categories of very unequal size, the smallest of which are why half of section 7 is unmeasurable.
- η and α are each fixed at one value. ε is swept in section 8.2, on a coarse grid that does not resolve where the constraint stops binding.
- The reliability threshold of 30 is a conventional rule of thumb. Wilson intervals are reported for every subgroup so a reader can apply their own.

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